EPFL

Laboratory of Intelligent Systems

SEMESTER PROJECT

Title: Body Rate Controller for new Avian Inspired Drone

Student(s): Antoine Isaac Adel Hinary (Robotics)

Professor: Dario Floreano

Assistant 1: Simon Jeger

Project description:

At the LIS we have developed a new AID, which for now only flies through manual control inputs. To be able to leverage all available degrees of freedom, we want to implement an existing controller on this new platform.

The goal of this project is to implement and tune a mapping-based PID controller on the new AID using the onboard Pixhawk framework.

Remarks:

You should present a research plan (Gantt chart) to your first assistant before the end of the second week of the project. An intermediate presentation of your project, containing 8 minutes of presentation and 7 minutes of discussion, will be held on 29 October 2024. The goal of this presentation is to briefly summarize the work done so far and discuss a precise plan for the remaining time of the project. Your final report should start by the original project description (this page) followed by a one page summary of your work. This summary (single sided A4), should contain the date, laboratory name, project title and type (semester project or master project) followed by the description of the project and 1 or 2 representative figures. In the report, importance will be given to the description of the experiments and to the obtained results. A preliminary version of your report should be given to your first assistant at the latest 10 days before the final hand-in deadline. A PDF (dated & signed) of your final version should be
Sent to your first assistant and to the administrative assistant of the lab before noon 3rd January 2025. Failing this deadline will reflect in the final grade. A 20 minute project defense, including 5 minutes for discussion, will take place 10 January 2025. You will be graded based on your results, report, final defense, and working style. All documents, including the report (source and pdf), summary page and presentations along with the source of your programs should be handed-in as a single compressed file on the day of the final defense at the latest.

Responsible professor:	Responsible assistant:
Signature: Dario Floreano	Signature: Simon Jeger

Lausanne, 02 September 2024

Semester Project Summary

Laboratory of Intelligent Systems (LIS)

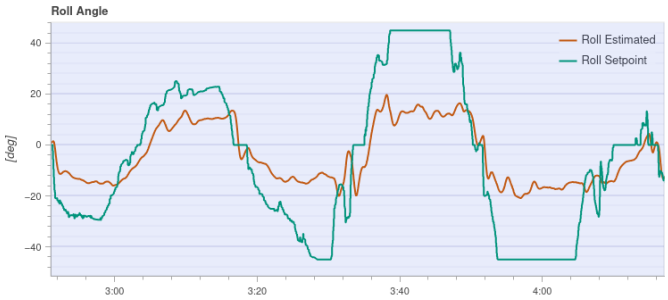
Ecole Polytechnique Fédérale de Lausanne (EPFL)

Project Title: Body Rate Controller for an Avian-Inspired Drone

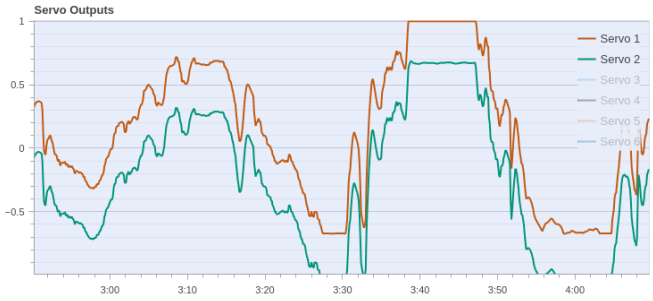
Date: January 2025

Project Summary: This semester project focuses on the design, implementation, and validation of a body rate controller for an avian-inspired drone. The primary objective was to integrate a dynamic actuator allocation system into the PX4 autopilot pipeline, leveraging a sensitivity matrix to adapt control efforts in real time. The project involved analyzing the aerodynamic behavior of morphing surfaces and their influence on the allocation matrix. While dynamic computations were partially integrated, the experimental results were based on a fixed allocation matrix at a 0 degree angle of attack. Key challenges addressed included memory optimization, debugging, and subsystem integration. Future efforts will aim to automate the dynamic allocation and improve computational designs. This research lays the foundation for advancing adaptive control strategies for morphing drones.

Important Results:



(a) Roll Angle Response



(b) Wing Sweeps

Fig. 1. (a) Roll angle response showing the estimated (orange) and setpoint (green) values over time. Deviations highlight aerodynamic and mechanical limitations during testing. (b) Wing sweep actuator response corresponding to roll movements, showing dynamic adjustments of the left (orange) and right (green) wing sweep actuators to achieve desired roll angles using the customized allocation matrix.

Body Rate Controller for an Avian-Inspired Drone

Antoine Hinary, Simon Jeger, Hoang-Vu Phan, Dario Floriano, *Fellow, IEEE,*

***Abstract*—Avian-inspired drones, with their morphing wing and tail surfaces, offer enhanced agility and energy efficiency in flight. However, fully leveraging these capabilities remains challenging due to the complex dynamics and control requirements of their morphing configurations. This work introduces a body-rate controller for an avian-inspired drone, designed to utilize all available actuators to achieve precise motion control to track attitude setpoints. Integrated into the PX4 pipeline, the controller eliminates the need for external computational units, enabling real-time and robust control. Experimental validation highlights the system's effectiveness in static conditions, with accurate tracking and adaptability to aerodynamic disturbances. This research establishes a foundation for advancing autonomous avian-inspired drones, emphasizing the role of morphing in achieving enhanced flight performance and energy efficiency under varied conditions without the need for an external Onboard Computer.**

***Index Terms*—Avian-inspired drones, body rate controller, PX4 autopilot, QGround Control, Effectiveness Matrix, Sensetvity Matrix.**

I. Introduction

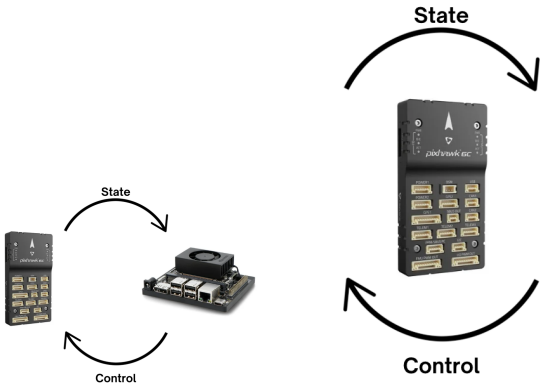
AVIAN-INSPIRED drones represent a significant advancement in aerial robotics, leveraging morphing surfaces to achieve high adaptability, agility, and energy efficiency. Morphing wings and tails enhance flight performance, with tail morphing improving stability and maneuverability [1], and coordinated wing-tail morphing demonstrating gains in energy efficiency and aerodynamic resilience [2]. Morphing technologies in wing designs have been extensively studied for their potential to improve aerodynamic performance and mission adaptability [3].

Controlling avian-inspired drones poses significant challenges due to the coupling effects of morphing wings and tails, requiring sophisticated strategies. Traditional control systems often rely on fixed allocation matrices, limiting adaptability to dynamic flight conditions. While the LisEagle drone has demonstrated dynamic control, it relies on a companion computer, such as a Jetson. Eliminating this external computational unit reduces weight and complexity, enabling us to move closer to the ultimate goal: performing diving maneuvers with the Lis-Raptor.

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(a) Avian Inspired Drone : Lis-Raptor



(b) Traditional control scheme (c) Proposed streamlined architecture

Fig. 2. Overview of the avian-inspired drone Lis-Raptor and control system architectures. (A) illustrates the Lis-Raptor drone used in this work, with a smaller wing area and lightweight, efficient hardware featuring 7 DOFs including (a propeller for thrust and 6 DOFs in controlled surfaces). (B) depicts the traditional control architecture with added computational and weight overhead due to the reliance on a companion computer. (C) shows the proposed streamlined architecture operating entirely on the Pixhawk 4, eliminating the need for external computational units while reducing system complexity and weight.

This work addresses these challenges by implementing a new body rate controller directly on the Pixhawk 4 autopilot platform. This is particularly critical for the Lis-Raptor, which has a smaller wing area, emphasizing

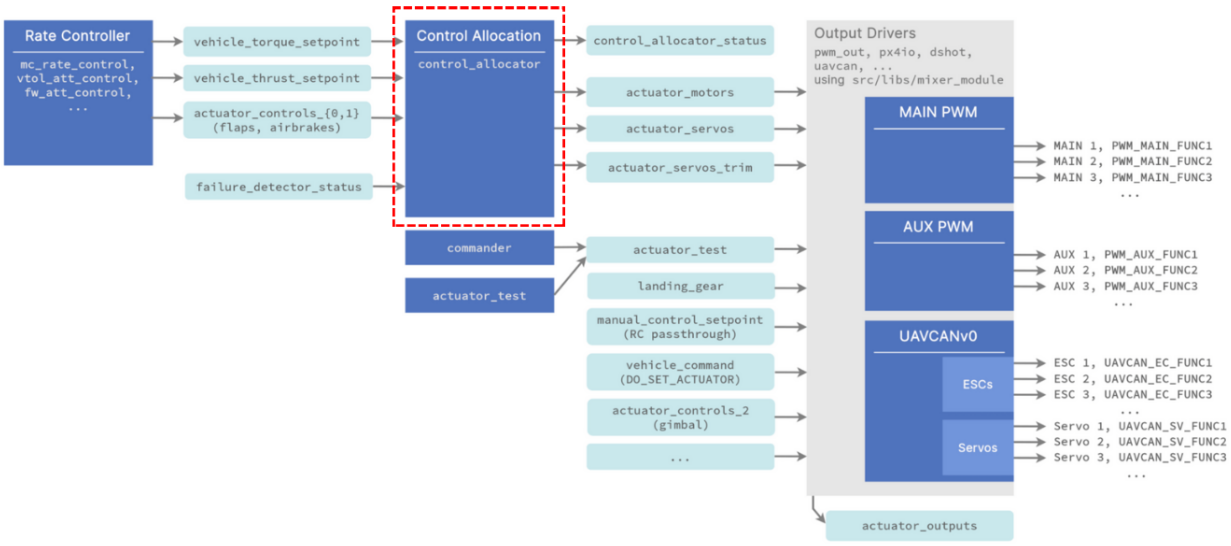


Fig. 3. Control architecture of the PX4 autopilot for fixed-wing aerial vehicles, showcasing the integration of the dynamic allocation matrix into the pipeline. The pipeline begins with the rate controller, which processes high-level commands (e.g., roll, pitch, and yaw setpoints) and generates control outputs, including torque and thrust setpoints. These outputs are sent to the control allocation module (highlighted with a red dashed line), where our dynamic allocation matrix is integrated. This module computes the actuator-specific commands based on the dynamically changing aerodynamic states and actuator effectiveness. Downstream, these actuator commands are distributed to output drivers, including PWM signals for servos and motors, ensuring precise control of the vehicle's aerodynamic surfaces and propulsion system. This architecture enables real-time control and adaptability, accommodating the morphing configurations of the Lis-Raptor drone.

the need for lightweight, efficient control systems. The controller integrates into the PX4 pipeline and employs flat plate theory for aerodynamic modeling. While simplified, flat plate theory effectively captures key aerodynamic behaviors and is suitable for high-angle-of-attack maneuvers [4], [5]. The system enables real-time dynamic allocation of control effort, streamlining the architecture to improve reliability and reduce system weight.

II. Method

A. PX4 Control Architecture and Dynamic Allocation System

The PX4 autopilot platform [6] employs a hierarchical control architecture tailored for fixed-wing aerial vehicles, designed to ensure stability and responsiveness under varying flight conditions. Similar advanced control architectures have been employed for agile fixed-wing UAVs, showcasing the effectiveness of cascaded controllers in managing complex flight dynamics [7]. The control framework integrates feed-forward elements, proportional-integral (PI) controllers, and airspeed-dependent scalers for precise actuator command generation.

The control process begins with the attitude vector Ψ, consisting of roll, pitch, and yaw angles in the inertial frame. High-level setpoints (Ψ_{sp}) are processed through proportional controllers to generate angular rate commands ($\dot{\Psi}_{sp}$). These angular rates are transformed to the body frame using the Jacobian matrix J_l^b, accounting for the dynamics of the vehicle.

In the body frame, the angular rate commands (ω_{sp}^b) are compared to the estimated angular rates ($\hat{\omega}^b$), and the resulting error is processed by a proportional-integral (PI) controller. The PI controller output is adjusted using airspeed-dependent scalers, ensuring that control outputs are appropriately adapted to varying flight conditions, such as changes in true airspeed (V_T) and indicated airspeed (V_I).

A feed-forward gain (FF) is applied to improve system responsiveness, and the scaled outputs are passed through a mixer, which computes the actuator commands (u) based on the control allocation matrix. The actuators, represented as δ, implement these commands, producing deflections that adjust the vehicle's aerodynamic surfaces.

This hierarchical approach, as depicted in Figure 4, ensures precise and robust control of the fixed-wing platform across a range of flight conditions by dynamically adjusting the control gains and actuator responses using airspeed information.

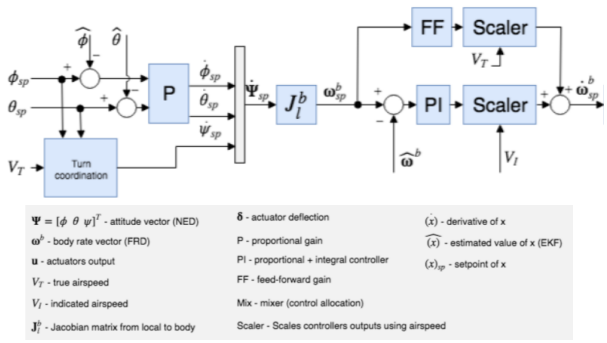


Fig. 4. PX4 fixed-wing Rate Controller Archetecture. The block diagram illustrates the hierarchical control framework, integrating feed-forward gains, proportional-integral (PI) controllers, and airspeed scalers to ensure precise and adaptive control. The legend provides details on the variables and components involved in the system, emphasizing the use of dynamic scaling and mixer allocation for actuator outputs.

While the main control logic was not altered, the standard PX4 setup relies on a static allocation matrix to map control commands to actuators. This approach is suitable for fixed-wing and multirotor systems but does not accommodate the dynamically changing aerodynamic properties of morphing drones. To address this limitation, a dynamic actuator allocation system was developed and integrated into the existing PX4 control pipeline, enhancing adaptability for the Lis-Raptor drone.

The dynamic allocation system computes real-time effectiveness matrices based on:

- Current actuator positions (e.g., wing sweep, tail pitch),
- Real-time vehicle attitude (roll, pitch, yaw).

This integration ensures that control efforts are distributed effectively, even during morphological adjustments, without altering the core control structure. The integration of the dynamic allocation module within the PX4 control architecture ensures compatibility with existing components, as shown in Figure 3.

B. Sensitivity Matrix Computation

A key component of the dynamic allocation system is the sensitivity matrix, which quantifies the influence of actuator states on aerodynamic outputs, guiding the control allocation process. This matrix was computed numerically by first simulating baseline aerodynamic responses. Then, small perturbations (Δx) were introduced to actuator states, and the normalized changes in aerodynamic outputs were calculated using the equation:

$$S = \frac{\Delta a}{\Delta x}. \tag{1}$$

The sensitivity matrix, illustrated in Figure 5, provided valuable insights into the effectiveness of each actuator, enabling informed adjustments to the dynamic allocation system and highlighting the relative contributions of individual actuators to aerodynamic outputs. For instance, right and left wing sweeps exhibit high sensitivity to roll, while tail sweep primarily affects pitch. This matrix underscores the varying impact of individual actuators on each aerodynamic response, guiding the design and adjustment of control allocation strategies to ensure optimal performance under diverse aerodynamic conditions.

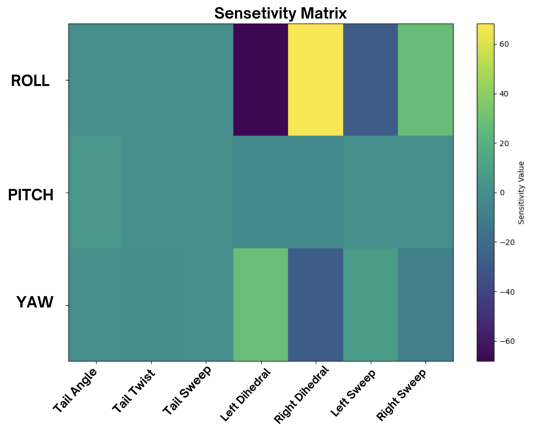


Fig. 5. Sensitivity matrix illustrating the influence of each actuator on the drone's aerodynamic outputs (roll, pitch, and yaw). The horizontal axis represents the actuators, including tail angle, tail twist, tail sweep, left and right dihedral, and left and right wing sweep. The vertical axis corresponds to the controlled body rates (roll, pitch, yaw). The color scale indicates the sensitivity value, where brighter colors represent higher sensitivity.

C. Aerodynamic Model

The drone's aerodynamic behavior is modeled using flat plate theory, which provides a simplified representation of aerodynamic forces and moments while capturing the key dynamics of a morphing drone. While flat plate theory is an approximation, it has been validated as an effective approach for modeling aerodynamic behaviors in scenarios involving high angles of attack and unsteady flow conditions [4], [5]. This makes it particularly suited for representing the morphing wing dynamics required for the Lis-Raptor's diving maneuvers. The six degrees of freedom (DOFs) correspond to the independent control of specific aerodynamic surfaces. Independent Wing Sweep enables asymmetrical configurations, significantly influencing roll and yaw control. Tail morphing incorporates multiple degrees of freedom, including Pitch, which controls the angle of attack (α) for vertical stability; Yaw which adjusts the horizontal position to optimize aerodynamic efficiency; Roll, which modifies tail twist for enhanced roll control; and Sweep, which changes the effective surface area (A) to manipulate lift and drag forces.

The aerodynamic forces and moments are computed using the flat plate model:

$$F_L = \frac{1}{2}\rho v^2 C_L(\alpha, \beta) A, \tag{2}$$

$$F_D = \frac{1}{2}\rho v^2 C_D(\alpha, \beta) A, \tag{3}$$

where:

- F_L and F_D are the lift and drag forces, respectively,
- ρ is the air density,
- v is the relative airflow velocity,
- C_L and C_D are the lift and drag coefficients, which depend on the angle of attack (α) and sideslip angle (β),
- A is the effective surface area of the aerodynamic component.

The aerodynamic moments are derived as:

$$M = r \times F, \tag{4}$$

where r is the distance vector from the center of mass to the point of force application.

The non-linear relationship between lift force, velocity, and angle of attack demonstrates that lift force remains relatively low at smaller angles of attack. However, it increases sharply beyond a critical threshold due to the combined effects of higher angles and airspeed. This behavior underscores the necessity of dynamically adjusting the allocation matrix to ensure optimal performance across varying aerodynamic conditions.

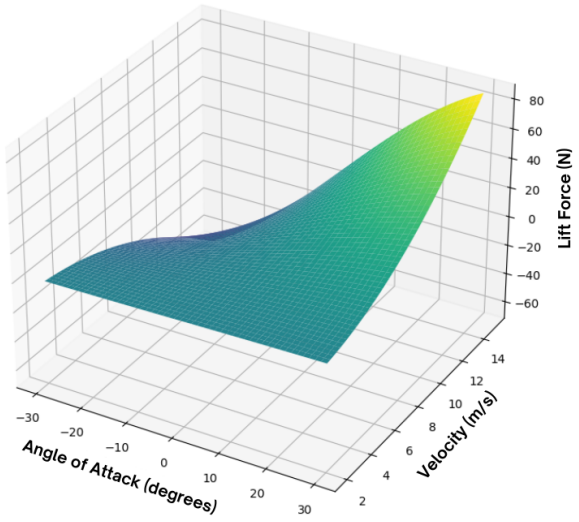


Fig. 6. Lift force variation as a function of velocity and angle of attack. The non-linear behavior at higher angles of attack and velocities emphasizes the need for a dynamic allocation matrix to optimize flight performance under varying aerodynamic conditions.

D. Actuator Dynamics

Each actuator is modeled as a first-order dynamic system with a time constant τ to approximate realistic response behavior. The system is represented by the transfer function:

$$H(s) = \frac{1}{\tau s + 1}, \tag{5}$$

where:

- $H(s)$ is the system's transfer function,
- τ is the time constant (set to 0.1 seconds in this model),
- s is the Laplace transform variable.

For the time-domain response, the relationship between the input command $u(t)$ and the actuator state $x(t)$ is given by:

$$\tau \frac{dx(t)}{dt} + x(t) = u(t). \tag{6}$$

This model ensures that the dynamic response of the actuators, including inherent delays, is realistically captured.

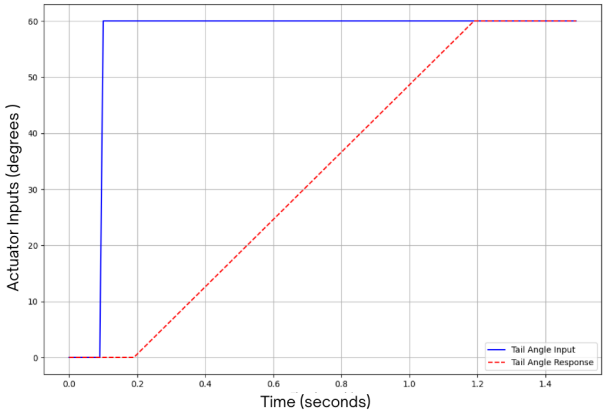


Fig. 7. Step response of the actuator model showing the input command (blue) and the output response (red). The results validate the accuracy of the first-order system in capturing realistic actuator dynamics, including inherent delays.

E. Allocation Matrix

The aerodynamic model and actuator dynamics are integrated into the control system to compute the effectiveness of control surfaces in real-time. This integration forms the basis of the dynamic allocation matrix, which is continuously updated during flight to account for changes in aerodynamic states caused by the drone's morphing configurations.

The dynamic allocation matrix assigns control effort to each actuator based on its current effectiveness, as determined by the aerodynamic conditions and the drone's state. This is achieved by calculating the sensitivity of

each actuator to aerodynamic forces and moments using the sensitivity matrix:

$$S = \frac{\Delta F}{\Delta x}, \tag{7}$$

where ΔF represents the change in aerodynamic forces or moments, and Δx represents the actuator displacement. The sensitivity matrix quantifies how small changes in actuator positions influence the aerodynamic outputs, serving as a key input to the allocation matrix.

By dynamically adapting the allocation matrix during flight, the control system can respond to varying aerodynamic conditions, such as changes in airspeed, angle of attack, or morphing states. This ensures that the Lis-Raptor drone maintains robust and efficient control performance, even under complex and rapidly changing flight scenarios.

III. Setup

The experimental setup consists of a wind tunnel in front of which the Lis-Raptor drone is suspended using an elastic cable capable of supporting up to 6 kg of weight. This configuration allows the drone to remain stable while enabling rotations around all axes, facilitating effective controller testing. In addition varying wind speeds have been used starting from 20% airspeed up to 60% airspeed.

A regulated power supply, positioned on the ground behind the drone, provides power to the drone via its power input. The voltage and current supplied to the drone can be adjusted to meet specific experimental requirements. This setup ensures consistent power delivery and allows precise control of the drone's operational parameters during testing.

To address mechanical constraints, two custom 3D-printed parts were developed for the testing setup. The first is an attachment mechanism structure designed to enable 3-axis rotation around the drone's center of mass, providing flexibility during experiments. The second is a protective component that prevents damage to the drone's foam skeleton by evenly distributing the tension from the suspension cord.

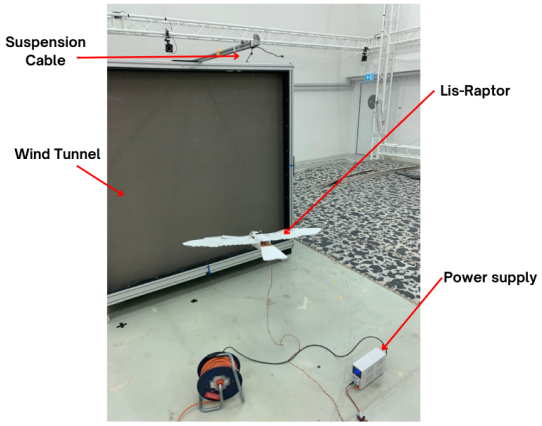


Fig. 8. Experimental setup showing the suspended Lis-Raptor drone in front of the wind tunnel, with the power supply visible in the background and suspended via a cable to a metal bar on top of the wind tunnel.

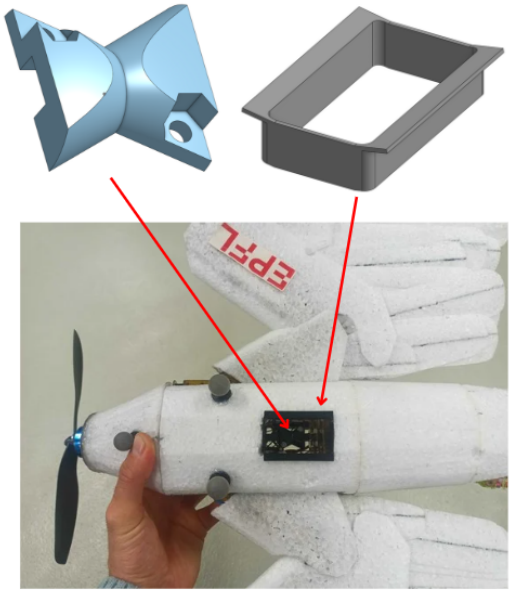


Fig. 9. 3D-printed components used for the setup and testing of the Lis-Raptor drone. The top-left image shows the attachment mechanism, allowing the drone to be suspended securely while enabling 3-axis rotation around its center of mass. The top-right image displays a protective component designed to shield the foam body of the drone from potential damage caused by the tension of the suspension cable. The bottom image illustrates the complete assembly of these components on the Lis-Raptor, ready for wind tunnel testing.

IV. Results

A. Experimental Validation

The Lis-Raptor drone's body rate controller was tested under static conditions by suspending the drone in front of a wind tunnel operating at 40% airspeed. During this experiment, attitude setpoints were manually sent

to the system in stabilized mode, and the corresponding responses were recorded for pitch and roll.

The roll angle response demonstrates closer adherence to the setpoint signal, though it is not entirely perfect. Similar to the pitch response, the limitations of the testing setup, particularly the attachment mechanism, constrained the drone's movements. While roll tracking performed better overall, these physical restrictions likely contributed to minor deviations, underscoring the influence of the experimental setup on control performance.



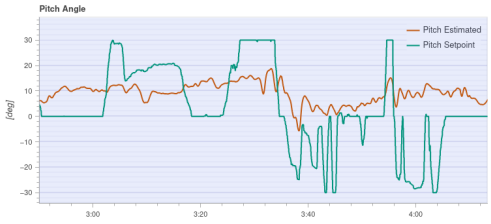
(a)



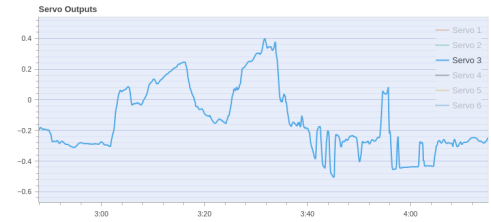
(b)

Fig. 10. (a) Roll angle response showing the estimated (orange) and setpoint (green) values over time. Deviations suggest mechanical and aerodynamic limitations during testing, but good tracking performance can be noticed. (b) Wing sweep actuator response, illustrating the dynamic adjustments made by the wing sweep actuators to achieve the desired roll angle during the experiment (in orange the left wing sweep and in green we have the right wing sweep).

The pitch angle response shows that the estimated pitch angle attempts to follow the manually provided setpoints but does so sub-optimally. This behavior can be attributed to specific testing mechanism limitations. Due to the drone's size and weight constraints, the 3D-printed components, such as the attachment mechanism and the protective part, were restricted to a compact design. Additionally, the hole made in the drone's foam body to allow for rotational freedom was necessarily small to maintain structural integrity. These size constraints collectively limited the range of motion for the drone during testing, directly impacting its ability to achieve the desired pitch adjustments. These parts were specifically designed to fit the drone's size and shape, which inherently restricted its range of motion. Furthermore, connecting the power supply from the back of the drone introduced an additional pitch moment due to the cable's weight and tension, making accurate pitch tracking more challenging.



(a)



(b)

Fig. 11. (a) Pitch angle response showing the estimated (orange) and setpoint (green) values over time. Deviations highlight aerodynamic and mechanical limitations in the setup, but initial tracking can be seen. (b) Tail actuator pitch response, corresponding to the elevator position (tail pitch), showing the actuator's attempts to adjust the pitch angle dynamically based on the setpoints provided.

V. Challenges in Dynamic Allocation Matrix Development

Implementing the dynamic allocation matrix within the PX4 framework presented a series of challenges, both anticipated and unforeseen. The primary goal was to ensure real-time adaptability of the allocation matrix to the drone's morphing configuration and flight conditions. Below, we outline the major challenges faced during this process, including the solutions implemented and unresolved issues.

A. Challenges Addressed

1) **Dynamic Computation of Allocation Matrix:** Traditional allocation matrices in PX4 are static and designed for fixed-wing or multirotor vehicles. Adapting this structure to a dynamically morphing drone required the development of a new Actuator-Effectiveness class capable of updating the matrix in real-time based on:
   - Actuator positions (e.g., wing sweep, tail pitch).
   - Vehicle attitude (roll, pitch, yaw).

   This was achieved by integrating the sensitivity matrix into the computation, allowing dynamic control allocation matrix under manual testing. However this uncovered other issues in our initial approach with unexpected challenges.

2) **Memory Management and Optimization:** The initial implementation suffered from memory inefficiencies due to frequent updates and computations to create the adequate matrices. This resulted in communication breakdowns between PX4 and QGroundControl. To address this, I optimized the memory allocation process and minimized redundant calculations, ensuring smoother real-time operation.

3) **Logging and Debugging Tools:** Debugging the dynamic allocation system was challenging due to the limitations of PX4_INFO for extensive logging in addition to not having access to a debugger tool. To overcome this, I implemented a custom uORB topic to log relevant data and modified the PX4 logger to record my new topic. This provided detailed insights into system behavior and allowed us to better understand the allocation matrix' behavior.

B. Unresolved Challenges

1) **Computational and Scaling Challenges:** While optimizations reduced memory usage, the computational overhead of frequent matrix updates remains high. This impacts the controller's ability to respond to rapid changes in aerodynamic conditions, particularly during high-speed maneuvers. Addressing these issues will require further optimization of the matrix computation and scaling processes.

2) **Scaling of Allocation Matrix Values:** Despite significant progress, scaling the dynamically computed allocation matrix to produce metric-consistent values remains unresolved. Incorrect scaling led to control allocation errors, due to the fact that the underlying control architecture assumes such normalization before computing and sending the control commands.

3) **Integration with Higher-Level Controllers:** The dynamic allocation system occasionally faced compatibility issues when interfacing with higher-level controllers (e.g., body rate controller). These issues stemmed from mismatches in expected input-output relationships between subsystems.

VI. Discussion and Conclusion

This work presents the design, implementation, and validation of a body rate controller for an avian-inspired drone, integrated into the PX4 autopilot pipeline. The controller incorporates a dynamic actuator allocation system, utilizing a sensitivity matrix to adapt control efforts in real time to changes in actuator positions and aerodynamic conditions. While this report demonstrates progress in integrating the dynamic computation of the allocation matrix within the PX4 pipeline, the implementation was limited to manual commands and did not operate while the PX4 system was armed due to software limitations. Consequently, the results presented were obtained using a fixed allocation matrix, precomputed with a pitch angle set to zero. This highlights the foundational groundwork laid for dynamic allocation, while emphasizing the need for further integration and automation to fully realize the potential of the system.

Experimental results demonstrated the system's potential for precise control of roll, pitch, and yaw in controlled environments. While the static allocation matrix provided stability, limitations were evident under dynamic scenarios. Pitch control was notably affected by mechanical constraints in the testing setup, including the attachment mechanism, foam protection component, and the additional pitch moment induced by the power supply cable. Roll tracking was closer to setpoints but also constrained by the physical design of the testing rig.

Unresolved challenges include computational overhead from frequent matrix updates, scaling inconsistencies in the allocation matrix, and occasional compatibility issues with higher-level controllers. These factors limited the system's responsiveness and accuracy during rapid changes in aerodynamic conditions or high-speed maneuvers.

Future efforts will focus on optimizing matrix computations, ensuring consistent scaling, and improving subsystem integration. Enhancements to the mechanical design of the testing setup will also address physical constraints observed during experiments. By addressing these challenges, the proposed system has the potential to significantly enhance the robustness and adaptability of morphing drones in complex, real-world conditions. While fully integrating the dynamic allocation system into the PX4 framework remains the ideal long-term goal, a more practical and immediate next step would involve incorporating a Raspberry Pi onboard. Acting as an intermediary, the Raspberry Pi could establish a bridge with the PX4 to enable dynamic updates to the allocation matrix, leveraging its computational capabilities to handle real-time adjustments effectively.

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